

Making Tool Choice Observable: A Receipt-Safe Diagnostic Study of VOI-Based Memory in a Toolized Text Environment

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Abstract

Tool-using language agents must often decide whether to act immediately or pay for information that may improve a later action. In multi-tool settings, this splits into two questions: whether to ask and which tool family to ask. This report extends the midway study of `voi_memory_v1`, a family-scoped memory variant tested in a cloneable Toy-TextMDP environment with `SCAN`, `VERIFY`, and `CHOOSE` actions. Since the midway report, I added scaled dual-battery runs, repaired reference generation, stricter receipt checks, and a v4 aggressive-stress task ecology designed to expose scan/verify conflict. The final receipt-safe v4 seed-7 battery does not support a stable which-tool improvement: the locked `voi_memory_v1` comparison slightly worsens choose-swallow and family-mismatch rates from reference to stress, and verify usage drops. Both reference and stress blocked-step audits route all blocked rows to task pressure. The main contribution is therefore not a positive tool-choice claim, but a controlled diagnostic ladder that makes the claim testable and rejects it cleanly under the current task ecology.

1 Introduction

Language agents are increasingly evaluated as systems that reason, call tools, receive observations, and revise behavior over time (Yao et al., 2023; Shinn et al., 2023; Liu et al., 2024; Jimenez et al., 2024). A central decision in these systems is epistemic: should the agent act now, or should it ask for more information? In a multi-tool setting, there is a second layer. If asking is useful, which information source should be queried?

The midway report narrowed the original broad proposal into a controlled diagnostic problem. Rather than claiming that memory already solved tool choice, it asked whether a family-scoped memory variant, `voi_memory_v1`, could make

a meaningful which-tool problem observable in a small, cloneable text environment. That report showed progress on a 10-task smoke batch: nonzero tool-choice opportunity appeared, but tool precision stayed at ceiling and blocked rows under `voi_memory_v1` still routed to task pressure. The safest claim was whether-to-ask progress, not which-tool improvement.

This final report extends that arc with the missing large-scale and confirmation evidence. I ran receipt-safe dual batteries, repaired reference packs when preflight failed, and tested a stronger v4 stress ecology. The result is a stable negative finding: under the current task profiles, `voi_memory_v1` does not improve the locked which-tool comparison. This is still useful. It turns a vague agent-memory story into a falsifiable diagnostic ladder with explicit claim boundaries, receipt gates, and route decisions.

The paper makes three contributions. First, it defines the task and memory machinery precisely enough for a classmate to inspect it. Second, it reports the post-midway experimental ladder, separating offline diagnostics from receipt-safe live evidence. Third, it shows that the current bottleneck is best explained by `task_profile` / `task_pressure`, not by an accepted need to edit the gate, memory, or selector.

The final report is organized around two research questions. RQ1 asks whether the stronger post-midway task ecologies expose enough live scan/verify conflict for a which-tool comparison to be meaningful. RQ2 asks whether `voi_memory_v1`, once evaluated only through receipt-safe batteries, improves that comparison. The answer to RQ1 is yes in a limited diagnostic sense: the v4 stress pack exposes wrong-tool surface in non-memory modes. The answer to RQ2 is no: the locked memory mode does not improve the final reference/stress comparison.

2 Method

Environment. ToyTextMDP is a finite-horizon text decision environment in the reinforcement-learning sense (Sutton and Barto, 2018). At each state x_t , the agent sees a hidden-answer task through text. It may choose immediately with CHOOSE actions, or pay a small cost for information using SCAN or VERIFY. SCAN is cheaper but can be ambiguous; VERIFY is more reliable and is sometimes necessary under the active task profiles. The environment supports exact cloning, so candidate action sequences can be evaluated in deterministic successor states.

Example step. In one v4 stress task, the hidden answer is A. The visible candidates include direct CHOOSE_A, a cheap SCAN clue that supports A at cost .02, and a higher-reliability VERIFY clue that also supports A at cost .05. The live `voi_memory_v1` policy chooses A immediately and receives reward, but oracle replay scores the best tool action as SCAN (.98) over VERIFY (.95). This row is therefore a task success but a tool-choice failure: it enters the tool-opportunity denominator and is counted as choose-swallow.

Agent pipeline. The agent first generates candidate plans, filters ask candidates, scores the remaining candidates, and executes one action. The expanded decision equation is:

$$\begin{aligned} C_t &= \text{Gen}(x_t, M_t) \\ \tilde{C}_t &= \text{Gate}(C_t, x_t, M_t) \\ a_t &= \text{Select}(\tilde{C}_t). \end{aligned}$$

Here C_t is the candidate set, M_t is retrieved memory, and a_t is the selected action. The gate is not a general selector. It only decides whether ASK candidates remain eligible before final arbitration. The selector then compares direct choice, scan plans, and verify plans using counterfactual scores plus structured penalties.

Memory and preflight. `voi_memory_v1` stores typed `decision_lesson` records. Each lesson contains a tool family, local scope, trigger condition, action bias, and provenance. Operationally, retrieval maps a matched lesson to a small score-level bias on family-tagged candidates before selection; it is a VOI-inspired control signal, not a learned estimator of information value. Retrieval is family-scoped: lessons about verify are meant to influence verify-relevant steps rather than all asking behavior. This is intentionally weaker than

Term	Operational meaning
Tool-choice opportunity	A step where oracle replay exposes a relevant epistemic tool family and multiple tool/choose alternatives are live.
Step-local preflight	Offline pack check that asks remain valuable after one-ask successor states, not only at the initial state.
<code>decision_lesson</code>	Typed memory record with family, scope, trigger, action bias, and provenance.
Choose-swallow	Rate at which a mode chooses directly despite an oracle tool-choice opportunity.
Negative inversion	A harmful case where <code>voi_memory_v1</code> is worse than the selector baseline on the stress pack.
Blocked-step routing	Diagnosis of blocked <code>voi_memory_v1</code> ask opportunities into task pressure, surfaced ask deficit, or hard cutoff.

Table 1: Key definitions added in response to midway feedback.

claiming robust semantic understanding; it tests whether structured textual control signals can become behaviorally consequential, in the spirit of agent-model and verbal-memory work (Andreas, 2022; Park et al., 2023; Shinn et al., 2023).

Step-local preflight is an offline guard before live execution. It checks that a generated pack contains successor states where asking has positive value and direct choice does not dominate almost everywhere. This prevents a trivial distribution where all modes look good because the correct behavior is almost always immediate choice.

Receipts and route decisions. The receipt layer is a reproducibility gate rather than an analysis tool. A run must satisfy `run_completed=true`, `jsonl_integrity=pass`, `task_id_pairing=pass`, and `safe_for_interpretation=true`. Only then can the diagnosis scripts read the JSONL, compare the selected family to the oracle family, and route the battery. Route A would continue confirmation, Route B is a stable negative or task-pressure review, and Route C is reserved for negative inversion that would justify layer-level diagnosis. In the final battery, the route is B.

3 Experiments

All experiments use internally generated tasks rather than a public benchmark. This limits external realism but makes attribution tight, un-

Receipt-Safe Tool-Choice Diagnostic Flow

Diagnosis is not interpreted before receipt gates pass.

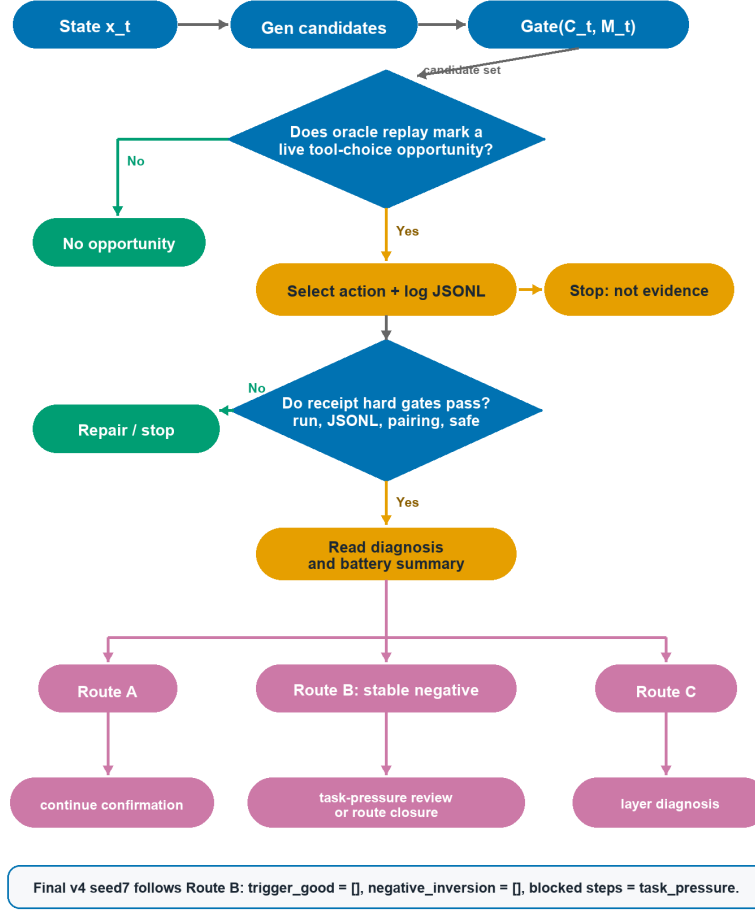


Figure 1: Receipt-gated decision pipeline. The gate is an ask-candidate filter, not the final selector, and diagnosis is interpreted only after the live JSONL passes the receipt hard gates.

like broader agent suites such as AgentBench and SWE-bench (Liu et al., 2024; Jimenez et al., 2024). Each live battery compares full (baseline), selector_penalty (a non-memory anti-ask variant), and voi_memory_v1 (family-scoped lessons). A battery is interpreted only if both packs pass four receipt gates: run completion, JSONL integrity, task-id pairing, and safety for interpretation.

Table 2 shows the post-midway ladder. The midway report ended at a 10-task smoke batch. The final project adds seed-7 v3 dual-battery evidence, a seed-11 stress-only probe, a repaired seed-11 reference confirmation battery, v4 offline repair, and a v4 seed-7 receipt-safe live battery.

The important methodological choice is that failures are preserved rather than overwritten. A reference pack that fails combined preflight is not frozen, hashed, or used in live evidence. A stress-only separation is not upgraded into a confirma-

tion result. This accounting is what makes the final negative result interpretable: the paper does not conclude “no effect” because one exploratory probe was weak; it concludes that the live evidence ladder repeatedly failed to turn stronger task pressure into a stable locked memory advantage.

4 Results and Analysis

The final v4 seed-7 battery is the current route authority. Both reference and stress packs pass the receipt layer. The battery summary routes to B: no trigger-good finding on the locked primary mode, no negative inversion, and a task-pressure/stable-negative review as the next action. This is a clean negative result, not a failed run.

The most important row is voi_memory_v1. In the reference pack, choose-swallow is 0.5291 and selected-vs-oracle family mismatch is 0.5291. In the v4 stress pack, these slightly worsen to 0.5376.

Post-Midway Evidence Ladder

Offline pack eligibility is separate from receipt-safe live evidence.

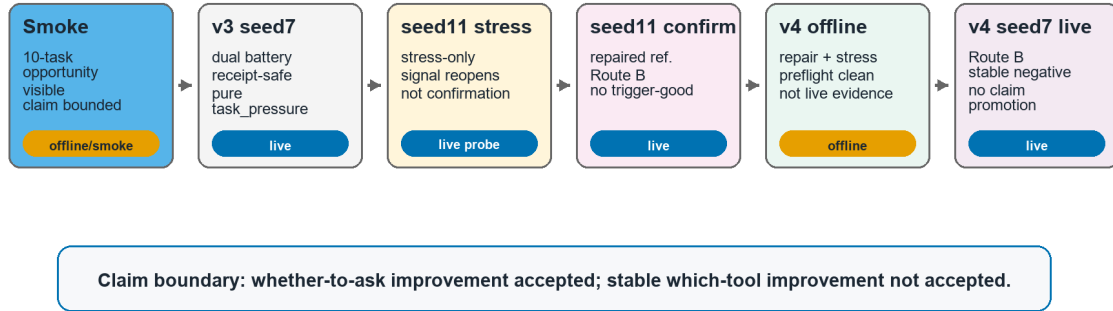


Figure 2: Post-midway diagnostic ladder. The visual separates offline pack eligibility from receipt-safe live evidence, ending in a Route B stable negative rather than claim promotion.

Stage	Seed/profile	Reference/stress status	Receipt status	Main outcome
Midway smoke	seed 7, 10-task step-local smoke verify_necessary		receipt checked	Nonzero opportunity, no accepted which-tool claim
v3 dual battery	seed 7, v3 stress	frozen reference vs v3 stress	both pass	Collapse on locked primary comparison; pure task pressure
seed-11 probe	seed 11, v3 stress	stress-only probe	stress pass	Stress-only family-mismatch separation reopens, but not confirmation
seed-11 confirmation	seed 11, repaired reference + v3 stress	repaired reference vs existing stress	both pass	Route B; no trigger-good; no negative inversion
v4 offline repair	seed 7, v4 stress	reference repair + regenerated stress	offline only	Both packs pass preflight; not live evidence
v4 live battery	seed 7, v4 stress	repaired reference vs v4 stress	both pass	Route B; stable task-pressure negative result

Table 2: Post-midway experimental ladder. Offline diagnostics are kept separate from receipt-safe live evidence.

Verify usage drops from 0.2063 to 0.0860. Thus the stronger v4 task ecology did not turn memory into a stable which-tool improvement. It instead preserved the same broad failure shape: the system often reaches tool-choice opportunities but still swallows them by choosing directly or by not using verify when the oracle says verify matters.

There are two useful secondary observations. First, the stress pack does expose wrong-tool surface in full and selector_penalty, where family mismatch rises above reference. This means the v4 stress design did create sharper scan/verify conflict. Second, that surface disappears in voi_memory_v1 not because verify succeeds, but because the mode still has high choose-swallow and lower verify usage. The negative result is therefore more informative than the midway smoke: the task ecology got sharper, but the locked memory mode still did not convert that pressure into reliable tool-family choice.

Figure 4 shows the distinction that motivated

the additional analysis. The denominator is not all environment steps; it is the subset where oracle replay says a tool family is relevant and the action space still contains meaningful tool/choose alternatives. Within that subset, direct choice is counted as choose-swallow, not as a tool-family success. This is why high opportunity counts can coexist with a negative route.

This is why the claim boundary matters. The accepted claim remains narrow: voi_memory_v1 improved whether-to-ask behavior in the earlier single-tool setting. The final batteries do not establish stable which-tool improvement, independent memory lift, or a reason to edit the ask gate, memory, or selector.

Error anatomy. The final failure is not a logging artifact and not simply a lack of task difficulty. The v4 stress pack does expose more wrong-tool surface in the non-memory modes, which means the generator successfully created more scan/verify conflict than the earlier smoke setting.

Pack	Mode	Tool	Oracle	Cov.	Swallow	Fam.	Verify	Wrong
Ref.	full	45	145	.310	.690	.690	.103	.000
	sel-pen	10	110	.091	.909	.909	.036	.000
	voi-mem	89	189	.471	.529	.529	.206	.000
Stress	full	51	151	.338	.662	.927	.298	.000
	sel-pen	21	121	.174	.826	.975	.174	.000
	voi-mem	86	186	.462	.538	.538	.086	.000

Table 3: Final v4 seed-7 battery metrics. Tool and Oracle are selected and oracle tool-choice step counts; Cov. is realized coverage. Swallow and Fam. mismatch are lower-better error rates; Verify is higher-better verify-use rate; Wrong is lower-better wrong-tool rate. The locked voi_memory_v1 reference-vs-stress comparison does not improve the which-tool metrics.

Final Locked Comparison: voi_memory_v1

Reference vs stress does not improve the which-tool metrics.

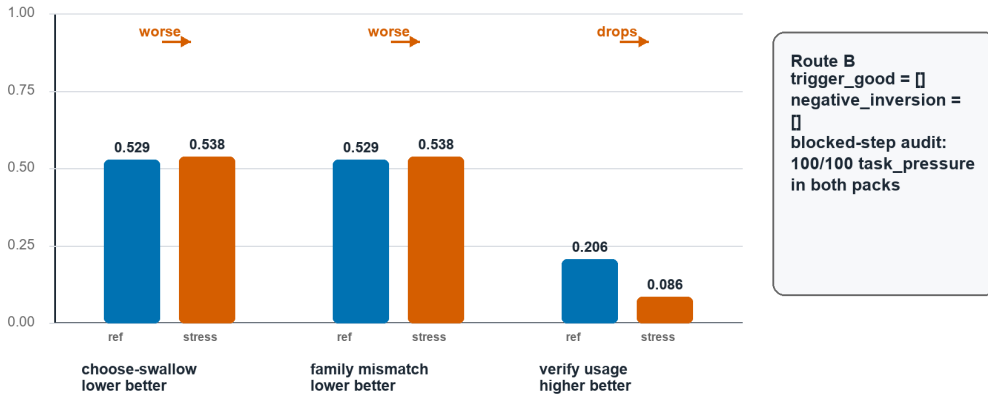


Figure 3: Final locked voi_memory_v1 comparison. Lower is better for choose-swallow and family mismatch, while higher is better for verify usage; the v4 stress pack does not improve the locked which-tool surface.

The locked memory mode, however, does not convert that exposed surface into verify usage. Its reference and stress family-mismatch rates remain nearly identical, and the stress-side verify usage falls below the reference side. The mechanism therefore looks like unrealized task pressure: the live policy sees opportunities but still fails to turn them into the oracle family action. This does not prove that the memory code is independently broken; it shows that, in this environment, structured memory bias is too weak or too poorly aligned to overcome direct-choice pressure.

Why Route B is the right conclusion. Route B is stricter than saying the experiment “did not work.” It means the run is valid, the JSONL can be interpreted, no negative inversion justifies a layer switch, and the locked primary comparison does not support a positive which-tool claim. This matters because the seed-11 stress-only probe briefly looked encouraging: family mismatch improved inside the stress pack. The repaired seed-11 ref-

erence confirmation and the v4 seed-7 live battery prevent that local pattern from being overpromoted. In other words, the framework rejects the claim for the same reason it could have accepted it: the comparison is now visible and receipt-safe.

Why not patch memory or selector now. The blocked-step audits route both final packs to task_pressure: 100 out of 100 blocked rows in the reference pack and 100 out of 100 in the stress pack. There is no surfaced ask deficit and no hard-cutoff candidate share. That does not prove memory and selector design are perfect. It says something narrower: the current receipts do not localize the failure to those layers. A selector or memory patch would be a new hypothesis about how to fight choose-swallow, not a conclusion licensed by this battery.

5 Discussion and Limitations

The project is small, but the negative result is disciplined. Many agent papers report headline scores

One Logged Step: Opportunity Without Success

Opportunity can exist even when the selected action fails the oracle tool family.

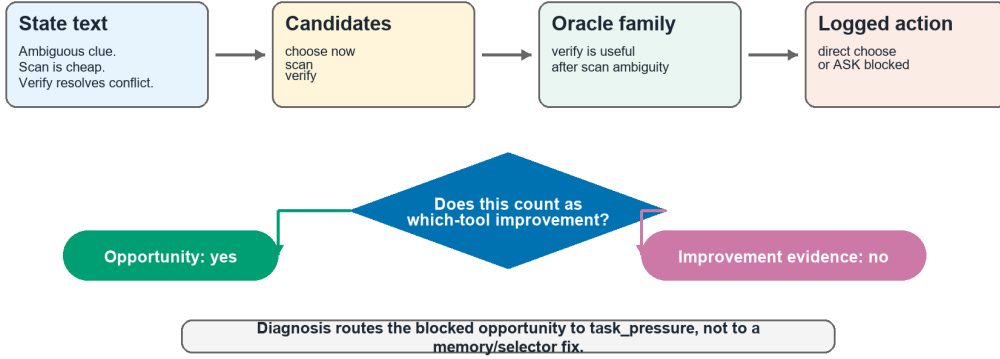


Figure 4: Case-step visualization. A step can count as a tool-choice opportunity while still failing the locked memory comparison because the selected action does not realize the oracle verify family.

on broad tasks; here the contribution is diagnostic control. The system can say when an offline pack is not eligible, when a live run is receipt-safe, when a stress-only signal is not enough, and when a route should stop. This style follows recent concerns that evaluation choices can change conclusions about system quality (Levtsov and Ustalov, 2025). For this project, the receipt and battery summary are not clerical details; they are the guardrails that prevent an attractive local pattern from becoming an unsupported claim.

The main limitation is external validity. Toy-TextMDP is synthetic and far smaller than open-ended code or web environments (Chen et al., 2021; Jimenez et al., 2024). Its advantage is exact replay and attribution, but the same control can make the state space pathological. For example, the v4 verify-usage drop could mean that `voi_memory_v1` failed, or that this micro-MDP created pressure where verify and scan are formally distinguishable but not semantically rich enough for a language-memory cue to matter. The safe interpretation is therefore local: receipt-safe negative evidence under this task ecology, not a universal claim that VOI memory cannot help tool choice.

A second limitation is the memory abstraction itself. The `decision_lesson` schema is structured text converted into candidate-family score bias. That is useful for a controlled mechanism test, but it is not a deep semantic model of information value. The negative result may therefore challenge the representation assumption as much as the task design. Future work should compare

this hand-structured memory against learned value predictors, paraphrase-stable triggers, and family-label swap tests before calling any positive effect semantic VOI reasoning.

The choose-swallow pattern also raises a broader agent-design question. If a policy repeatedly chooses directly despite receipt-safe tool-choice opportunities, natural-language penalties and discrete recalled lessons may be too weak to counter an immediate-generation prior. The present study cannot settle that question, because no negative inversion or surfaced ask deficit justifies a layer switch. It does, however, make the failure observable enough to test stronger anti-swallow designs as new hypotheses rather than retroactive explanations.

This version incorporates the midway feedback by defining the project terms operationally, decomposing the pipeline into generation, gating, selection, receipt, and diagnosis, and separating “opportunity exists” from “which-tool improvement is supported.”

The next scientific step is not to patch the gate or selector immediately. The current evidence points to either closing the present route as a stable negative result, or redesigning task pressure with even more live-aligned oracle exposure. A layer switch should require negative inversion or direct layer evidence, neither of which appears in the final battery.

6 Conclusion

Structured memory did not improve the locked which-tool comparison under the current task

ecologies. The contribution is a receipt-safe diagnostic ladder: it exposes tool-choice opportunities, rejects an unsupported positive claim, and points the next review toward task pressure rather than gate, memory, or selector changes.

Author Contributions

Bochuan Yao designed and implemented the toolized environment, VOI memory variants, task profiles, live batteries, diagnosis pipeline, and report analysis.

Code and Artifacts

The public code and artifact repository is <https://github.com/yaobc77-ai/cs505-playground-final-artifacts>.

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